

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- Enumerate three Collection interfaces suitable for this system.
- Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- Write a Java program using HashSet and HashMap with Generics to:
 - Add students
 - Map students to courses
 - Display data using Iterator

ANSWER:

Collection interfaces:

- Set → Ensures uniqueness of students. Since no two students can be duplicated in the system, a **Set** is ideal for storing student records.
- List → Useful for maintaining ordered lists of courses. For example, if a student enrolls in multiple courses, the order of enrollment can be preserved.
- Map → Provides key-value mapping. A student can be the key, and their enrolled courses can be the value, allowing efficient lookups.

Structure:

A **HashSet<Student>** ensures unique students. A **HashMap<Student, HashSet<Course>>** maps each student to their enrolled courses. This combination guarantees uniqueness and efficient retrieval.

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QUESTIONS

Q1 ✓
Student Registration System
10 marks

Q2 ✓
E-Commerce Cart System
10 marks

Q3 ✓
Library Management System
10 marks

Q4 ✓
Employee Data Processing
10 marks

Q5 ✓
Online Voting System
10 marks

5/5 answered

Q1 Student Registration System

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Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- (a) Enumerate three Collection interfaces suitable for this system.
- (b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- (c) Write a Java program using HashSet and HashMap with Generics to:
 - Add students
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 - Display data using Iterator

Your Code

```
1 import java.util.HashMap;
2 import java.util.HashSet;
3 import java.util.Iterator;
4 import java.util.Map;
5 import java.util.Set;
6 public class StudentRegistrationSystem {
7     public static void main(String[] args) {
8         Set<Student> students = new HashSet<>();
9         Map<Student, Set<String>> studentCourses = new HashMap<>();
10        Student s1 = new Student("S101", "Moshika");
```

Input

stdin input

Output

```
11      Student s2 = new Student("S102", "Arun");
12      Student s3 = new Student("S103", "Karthik");
13      students.add(s1);
14      students.add(s2);
15      students.add(s3);
16      studentCourses.put(s1, new HashSet<>());
17      studentCourses.put(s2, new HashSet<>());
18      studentCourses.put(s3, new HashSet<>());
19      studentCourses.get(s1).add("CO2");
20      studentCourses.get(s1).add("DSA");
21      studentCourses.get(s1).add("JAVA");
22      studentCourses.get(s2).add("CO2");
23      studentCourses.get(s2).add("DBMS");
24      studentCourses.get(s3).add("JAVA");
25      studentCourses.get(s3).add("OS");
26      Iterator<Student> studentIterator = students.iterator();
27      while (studentIterator.hasNext()) {
28          Student student = studentIterator.next();
29          System.out.println("Student: " + student);
30          Set<String> courses = studentCourses.get(student);
31          Iterator<String> courseIterator = courses.iterator();
32          System.out.print("Enrolled Courses: ");
33          while (courseIterator.hasNext()) {
34              System.out.print(courseIterator.next());
35              if (courseIterator.hasNext()) System.out.print(", ");
36          }
37          System.out.println("\n");
38      }
39  }
```

```
36         }
37         System.out.println("\n");
38     }
39 }
40 }
41 class Student {
42     private String id;
43     private String name;
44     public Student(String id, String name) {
45         this.id = id;
46         this.name = name;
47     }
48     public boolean equals(Object obj) {
49         if (this == obj) return true;
50         if (obj == null || getClass() != obj.getClass()) return false;
51         Student student = (Student) obj;
52         return id.equals(student.id);
53     }
54     public int hashCode() {
55         return id.hashCode();
56     }
57     public String toString() {
58         return id + " - " + name;
59     }
60 }
```

**Run****Save**

Output

Student: S102 - Arun

Enrolled Courses: CO2, DBMS

Student: S103 - Karthik

Enrolled Courses: JAVA, OS

Output

```
Enrolled Courses: JAVA, OS
```

```
Student: S101 - Moshika
```

```
Enrolled Courses: JAVA, DSA, CO2
```

Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- (a) List three Collection classes applicable for cart management.
- (b) Represent how cart items are stored using a List-based structure.
- (c) Write a Java program using ArrayList and Iterator to:
 - Add items
 - Remove an item
 - Display all items

ANSWER:

Collection classes:

1. ArrayList → Provides dynamic resizing and fast random access, making it suitable for cart operations.
2. LinkedList → Efficient for frequent insertions and deletions, useful when customers frequently update their cart.
3. Vector → Synchronized version of ArrayList, useful in multi-threaded environments where multiple users may access carts simultaneously.

Structure:

A `List<Product>` (commonly `ArrayList`) stores cart items in order. Each product can be added, removed, or displayed sequentially.

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5/5 answered

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- (c) Write a Java program using ArrayList and Iterator to:
 - Add items
 - Remove an item
 - Display all items

Your Code

```
1 import java.util.*;
2 public class EcommerceCart {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<String> cart = new ArrayList<>();
6         int n = sc.nextInt();
7         sc.nextLine();
8         for (int i = 0; i < n; i++) {
9             cart.add(sc.nextLine());
10        }
```

Input

```
3
Laptop
Mouse
Keyboard
Mouse
```

Output

```
Cart: [Laptop , Mouse, Keyboard]
Cart Items After removing an item:
Laptop
Keyboard
```

Q4 ✓
Employee Data Processing
10 marks

Q5 ✓
Online Voting System
10 marks

5/5 answered

Your Code

```
1 import java.util.*;
2 public class EcommerceCart {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<String> cart = new ArrayList<>();
6         int n = sc.nextInt();
7         sc.nextLine();
8         for (int i = 0; i < n; i++) {
9             cart.add(sc.nextLine());
10        }
11        System.out.println("Cart: " + cart);
12        String item = sc.nextLine();
13        for (int i = 0; i < cart.size(); i++) {
14            if (cart.get(i).equalsIgnoreCase(item)) {
15                cart.set(i, "");
16            }
17        }
18        System.out.println("Cart Items After removing an item:");
19        Iterator<String> it = cart.iterator();
20        while (it.hasNext()) {
21            String s = it.next();
22            if (!s.equals(""))
23                System.out.println(s);
24        }
25    }
26 }
```

Input

```
3
Laptop
Mouse
Keyboard
Mouse
```

Output

```
Cart: [Laptop , Mouse, Keyboard]
Cart Items After removing an item:
Laptop
Keyboard
```

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- (a) Name three Map implementations in Java.
- (b) Design a schema-like representation using Map for storing ISBN and book details.
- (c) Write a Java program using HashMap to:
 - Insert book details
 - Retrieve a book
 - Display all books

ANSWER:

Map implementations:

1. HashMap → Provides constant-time performance for insert and lookup, ideal for ISBN-based retrieval.
2. TreeMap → Maintains sorted order of keys, useful if books need to be displayed in ascending ISBN order.
3. LinkedHashMap → Maintains insertion order, useful for displaying books in the order they were added.

Schema:

`HashMap<String, Book>` where the key is ISBN (unique identifier) and the value is the book object containing details like title, author, and publisher.

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Q3 Library Management System

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- (c) Write a Java program using HashMap to:
 - Insert book details
 - Retrieve a book
 - Display all books

Your Code

```
1 import java.util.*;  
2 public class LibraryManagement {  
3     public static void main(String[] args) {  
4         Scanner sc = new Scanner(System.in);  
5         HashMap<Integer, String> library = new HashMap<>();  
6         int n = sc.nextInt();  
7         sc.nextLine();  
8         for (int i = 0; i < n; i++) {  
9             int isbn = sc.nextInt();  
10            sc.nextLine();
```

Input

```
3  
101  
Java  
102
```

Output

```
Retrieved Book: Python
```

```
Library Books:
```

Q4

Employee Data Processing

10 marks

**Q5**

Online Voting System

10 marks



5/5 answered

Your Code

```
1 import java.util.*;
2 public class LibraryManagement {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         HashMap<Integer, String> library = new HashMap<>();
6         int n = sc.nextInt();
7         sc.nextLine();
8         for (int i = 0; i < n; i++) {
9             int isbn = sc.nextInt();
10            sc.nextLine();
11            String book = sc.nextLine();
12            library.put(isbn, book);
13        }
14        int search = sc.nextInt();
15        if (library.containsKey(search))
16            System.out.println("Retrieved Book: " + library.get(search));
17        else
18            System.out.println("Book Not Found");
19        System.out.println("\nLibrary Books:");
20        for (Map.Entry<Integer, String> e : library.entrySet()) {
21            System.out.println(e.getKey() + " : " + e.getValue());
22        }
23    }
24 }
```

**Run****Save**

Input

Python

103

C++

102

Output

Retrieved Book: Python

Library Books:

101 : Java

102 : Python

103 : C++

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
- (b) Represent how employee data is stored using List with Generics.
- (c) Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

ANSWER:

Advantages of Generics:

1. **Type Safety** → Prevents runtime errors by ensuring only specific types are stored in collections.
2. **No Casting Required** → Eliminates the need for explicit type casting when retrieving elements.
3. **Code Reusability** → Allows writing generic classes and methods that work with different data types.

Structure:

`List<Employee>` ensures that only employee objects are stored, avoiding accidental insertion of unrelated data.

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5/5 answered

Q4 Employee Data Processing

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- (c) Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.List;
4 public class Main {
5     public static void main(String[] args) {
6         List<Employee> employees = new ArrayList<>();
7         employees.add(new Employee(101, "Asha", 45000));
8         employees.add(new Employee(102, "Karthik", 65000));
9         employees.add(new Employee(103, "Mohan", 52000));
10        employees.add(new Employee(104, "Priya", 30000));
11        Iterator<Employee> it = employees.iterator();
12        System.out.println("Employees with salary > 50,000:");
```

Input

stdin input

Output

```
Employees with salary > 50,000:
Employee{id=102, name='Karthik',
salary=65000.0}
Employee{id=103, name='Mohan',
salary=52000.0}
```



Online Voting System
10 marks

5/5 answered

```
6      List<Employee> employees = new ArrayList<>();
7      employees.add(new Employee(101, "Asha", 45000));
8      employees.add(new Employee(102, "Karthik", 65000));
9      employees.add(new Employee(103, "Mohan", 52000));
10     employees.add(new Employee(104, "Priya", 30000));
11     Iterator<Employee> it = employees.iterator();
12     System.out.println("Employees with salary > 50,000:");
13     while (it.hasNext()) {
14         Employee emp = it.next();
15         if (emp.salary > 50000) {
16             System.out.println(emp);
17         }
18     }
19 }
20 }
21 class Employee {
22     int id;
23     String name;
24     double salary;
25     Employee(int id, String name, double salary) {
26         this.id = id;
27         this.name = name;
28         this.salary = salary;
29     }
30     public String toString() {
31         return "Employee{id=" + id + ", name='" + name + "', salary=" +
32     }
33 }
```

Output

```
Employees with salary > 50,000:
Employee{id=102, name='Karthik',
salary=65000.0}
Employee{id=103, name='Mohan',
salary=52000.0}
```

Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- (a) Identify three suitable Collection types for this system.
- (b) Design a structure using Set and Map for voters and vote counting.
- (c) Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting • Display results

ANSWER:

Collection types:

1. HashSet → Ensures unique voters, preventing duplicate voting.
2. HashMap → Stores candidates as keys and vote counts as values, allowing efficient tallying.
3. TreeMap → Provides sorted results, useful for displaying vote counts in order.

Structure:

`HashSet<Voter>` ensures each voter votes only once. `HashMap<Candidate, Integer>` stores vote counts.

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- (b) Design a structure using Set and Map for voters and vote counting.
- (c) Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting • Display results

Your Code

```
1 import java.util.*;
2 public class OnlineVotingSystem {
3     private static Set<String> votedVoters = new HashSet<>();
4     private static Map<String, Integer> voteCounts = new HashMap<>();
5     public static void main(String[] args) {
6         String[] candidates = {"Alice", "Bob", "Charlie"};
7         for (String c : candidates) {
8             voteCounts.put(c, 0);
9         }
10        castVote("V1001", "Alice");
11        castVote("V1002", "Bob");
12        castVote("V1003", "Alice");
```

Input

stdin input

Output

```
Vote recorded: V1001 -> Alice
Vote recorded: V1002 -> Bob
Vote recorded: V1003 -> Alice
Duplicate vote rejected for voter:
V1001
```



Online Voting System
10 marks

5/5 answered

```
5 public static void main(String[] args) {
6     String[] candidates = {"Alice", "Bob", "Charlie"};
7     for (String c : candidates) {
8         voteCounts.put(c, 0);
9     }
10    castVote("V1001", "Alice");
11    castVote("V1002", "Bob");
12    castVote("V1003", "Alice");
13    castVote("V1001", "Charlie");
14    displayResults();
15 }
16 private static void castVote(String voterId, String candidate) {
17     if (votedVoters.contains(voterId)) {
18         System.out.println("Duplicate vote rejected for voter: " +
19             voterId);
20         return;
21     }
22     votedVoters.add(voterId);
23     voteCounts.putIfAbsent(candidate, 0);
24     voteCounts.put(candidate, voteCounts.get(candidate) + 1);
25     System.out.println("Vote recorded: " + voterId + " -> " + candidate);
26 }
27 private static void displayResults() {
28     System.out.println("\n--- Voting Results ---");
29     for (Map.Entry<String, Integer> entry : voteCounts.entrySet())
30         System.out.println(entry.getKey() + " : " + entry.getValue());
31 }
32 }
```

Output

```
Vote recorded: V1001 -> Alice
Vote recorded: V1002 -> Bob
Vote recorded: V1003 -> Alice
Duplicate vote rejected for voter:
V1001
```